

Development and Validation of an Instrumented Static Test Bench for Brushless Motors

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Abstract—The characterization of UAV propulsion systems requires reliable measurements of thrust and electrical performance, which are not always available from manufacturer datasheets under practical operating conditions. This paper presents the development and experimental validation of a low-cost instrumented static test bench for brushless motors used in UAV propulsion systems. The platform measures thrust, rotational speed, voltage, and current, enabling the analysis of propulsion performance and electrical power consumption. The mechanical structure was designed in CAD and fabricated using 3D printing with linear guides to reduce friction during thrust measurement. Sensor data are acquired using an Arduino-based system and processed in MATLAB for real-time visualization and analysis. Experimental validation was conducted using a Holybro motor-propeller assembly with reference data provided by the manufacturer. The results showed strong agreement with the expected propulsion behavior, confirming the quadratic relationship between thrust and RPM and the cubic trend between current and rotational speed. The proposed platform provides a reliable and accessible tool for propulsion testing and comparison of UAV motor-propeller configurations.

Index Terms—UAS Testbeds, Smart Sensors, Technology Challenges, Experimental Validation.

I. INTRODUCTION

The use of UAVs has become increasingly popular worldwide for a variety of applications, such as logistics, mapping, research, and agriculture. According to the DJI industry report, approximately 400,000 drones were used across 100 different countries for agricultural purposes in 2024 [1], and the advantages of using them are evident, especially in the context of current challenges in modern agriculture, such as global warming. The use of cameras and sensors on UAVs allows for the collection of data and the generation of precise analyses in a short time, identifying problems that would be much more difficult to detect with field inspections alone [2]. Beyond these applications, in Switzerland, for example, where there are 26 cantons, more than half of the cantonal police agencies use UAV's for security purposes [3]. It is also a method that enables a more advanced and safe study of wildlife [4], illustrating the expansion of UAV applications across different areas.

This increase in UAV usage intensifies the need for predictable performance and ever greater efficiency, which requires careful selection of components. In particular, selecting a motor-propeller assembly depends on reliable propul-

sion data, since thrust capability and the UAV's electrical power demand directly influence payload capacity, as well as ESC and battery selection. However, performance data is not always readily available. In practice, the propulsion data reported in the datasheet may be obtained under different conditions compared to the required ones or may omit important variables. Moreover, commissioning specific tests from the manufacturer can also be very expensive. Therefore, since the choice of the motor-propeller depends heavily on accurate performance data, this motivates the need for independent measurements capable of obtaining coherent and consistent results.

Previous studies demonstrate the relevance of experimental propulsion characterization for UAV design, although available test solutions often follow different design priorities than those adopted in this work. Deters [5] shows a configuration for static laboratory testing of small propulsion systems based on instrumentation focused on thrust and torque measurement, as well as automated data acquisition. Avanzini [6] developed a test bench capable of measuring thrust, shaft torque, electrical parameters, and track speed, using an aluminum structure and a more instrumented architecture. Piljek [7] proposed a characterization methodology based on a commercial dynamometer to generate static propulsion maps for multi-rotor electric propulsion units. These studies are important and interesting references. However, they do not focus on a low-cost, predominantly 3D-printed test bench built with readily available components and intended for direct replication. In this context, the present work addresses this need; a low-cost static test bench for brushless motors used in UAVs has been developed with the aim of measuring propulsion parameters. The system measures thrust, motor rotational speed (RPM), current, and voltage, allowing for the estimation of electrical power and load capacity, as well as the selection of ESCs and batteries.

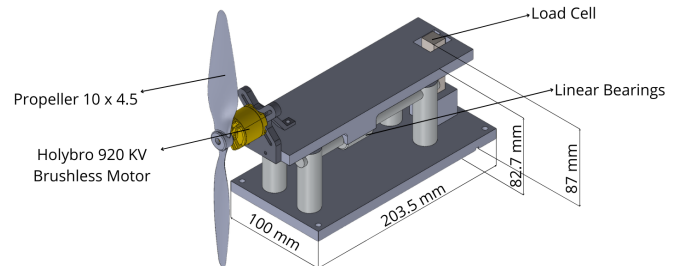


Fig. 1: CAD overview of the static test bench.

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II. MATERIALS AND METHODS

A. Mechanical Design and Assembly

The structural model of the test bench was entirely developed in Solidworks software, as shown in Figure 1. The assembly consists of a fixed lower base and four columns fitted into it, which are responsible for supporting the two linear bearing rods that have 12 mm diameter. The linear bearings move along these rods and are attached to the upper base, allowing the assembly to move. The upper base also has a hole for the passage of the load cell, and the lower base includes a reinforced support for its mounting and securing.

The test bench was mostly 3D printed using polylactic acid (PLA). A 40% infill was adopted to provide sufficient mechanical resistance during static tests while limiting printing time and material consumption. In addition, aluminum linear bearings were acquired to reduce vibrations, aiming to obtain more accurate results during the tests.

Figure 2 represents the operating principle of the test bench. The thrust force generated by the motor-propeller assembly is transmitted to the system through the aluminum linear bearings, allowing movement along the thrust axis. This displacement applies a force to the upper part of the load cell, which is subjected to shear stress, generating deformation measured by the strain gauges and converted into an electrical signal through the Wheatstone bridge. Since the resulting signal is typically low in amplitude, an HX711 module was used to amplify and digitize it, enabling it to be read by a microcontroller.

The load cell was also calibrated using the HX711-ADC library in the Arduino IDE software prior to testing. This procedure is necessary since the electrical readings of the system are obtained in converter units and not directly in grams, which is the desired unit. Thus, a known load was applied to the load cell, and the corresponding reading was recorded as a calibration reference, allowing for accurate readings in the correct unit.

Before fabricating the test bench, a static structural simu-

lation of the motor support was performed in SolidWorks to verify its integrity, considering the motor mount part as the most critical element, as it directly receives the thrust force from the motor-propeller assembly. In the simulation, the mounting holes of the support to the test bench were defined as constraints, represented by dashed green circles in Figure 3, while the load was applied to the motor coupling region along the propulsion axis, represented by the dashed red circle in the same figure. The applied force was defined based on the maximum thrust reported by the manufacturer for the reference assembly, converted to newtons and multiplied by a safety factor (FS) in order to guarantee a conservative operating condition. The mesh was locally refined in the regions of the holes and in the load application area, aiming for greater accuracy in stress estimation. The results of this step served as a safety criterion to proceed with the development of the test bench.

For the experiments, a Holybro 920 KV motor with a maximum power of 272 W and a 10×4.5 inch propeller (260×30 , mm) was selected. This setup was chosen because the manufacturer provides experimental data obtained under specific conditions, allowing the system to be assembled under equivalent conditions and used as a reference for validating the test bench. In addition, the electronic speed controller (ESC) must be selected adequately, since if the current required by the motor-propeller assembly exceeds the ESC capacity, it may suffer from excessive heating or the activation of the protection cutoff (power reduction or shutdown). Thus, a 20 A nominal-current ESC (BLHeli S firmware) was used, considering that the expected maximum current is 17 A, which provides a safe operating margin. A 4 cell 5200 mAh LiPo battery rated at 50C continuous discharge was also used during the tests.

As can be seen in Figure 2, the propeller radius is greater than the distance between the propeller center and the lower base. Thus, if the test bench were fully supported on a surface, the propeller would collide with it. To avoid this interference, the edge of the lower base was positioned at

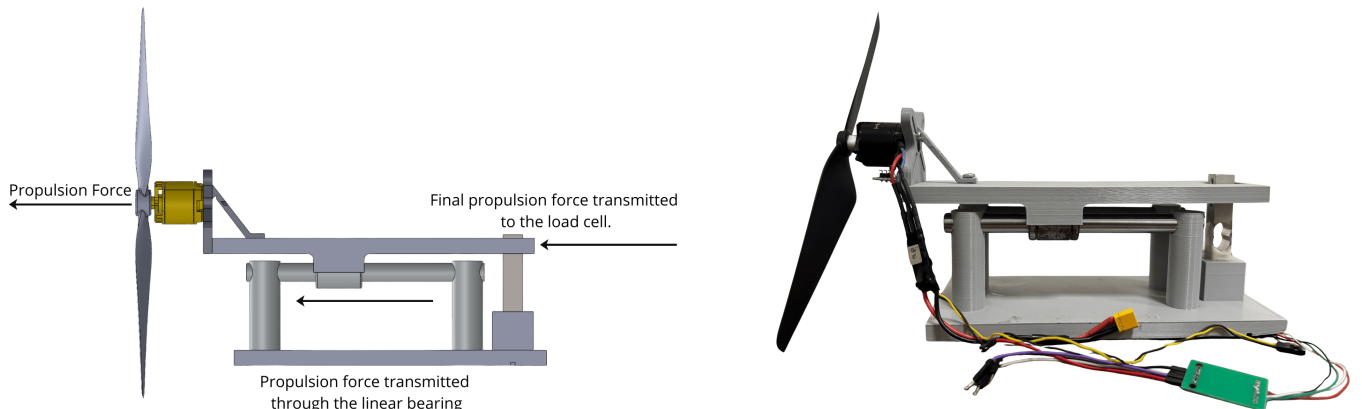


Fig. 2: Overview of the assembled 3D-printed test bench and its operating principle. The setup includes the main components such as the brushless motor and propeller, load cell, KY-003 Hall-effect sensor, ESC, HX711 load-cell amplifier, and linear bearings. The figure also illustrates the thrust transmission path from the motor-propeller assembly to the load cell through the carriage guided by linear bearings.

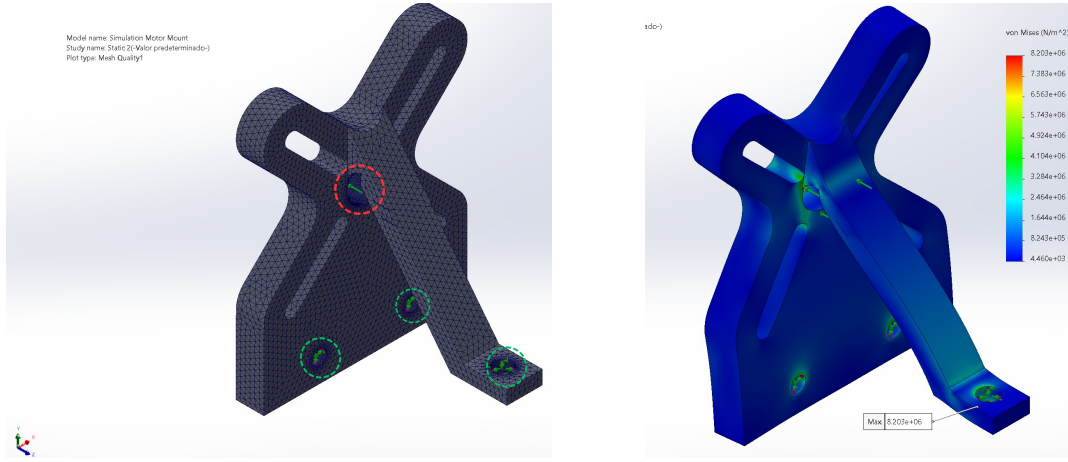


Fig. 3: Finite-element analysis of the motor mount in the SolidWorks Simulation environment: (a) deformation result showing the von Mises stress distribution; the displacement field is exaggerated 62 times for visualization purposes; (b) mesh control and boundary conditions used in the model, where green dashed circles indicate fixed regions and the red dashed circle indicates the force application area.

the edge of the table, leaving the region under the propeller free and allowing its rotation as well as the displacement of the upper base along the thrust axis.

During the tests, the lower base was clamped using C-clamps, which allow for quick assembly and disassembly, enabling the transportation and repetition of the tests at different locations. Alternatively, for permanent installations and to maximize the rigidity of the test bench, it can also be secured with bolts through the four holes in the lower base.

B. Electrical Schematics

The data acquisition system was designed to simultaneously obtain the thrust, current, voltage, and rotation speed (RPM) of the motor during testing. For this purpose, the dedicated electrical schematic was developed, as shown in Figure 4, in which the first three parameters mentioned above were measured as a function of RPM, prioritizing low cost and compatibility with the Arduino UNO microcontroller. The main objective of collecting this data is to compare it with the supplier's data, aiming to validate the test bench.

TABLE I: Components referenced in Fig. 4

No.	Component
1	Load cell YZC133
2	HX711 load cell amplifier (24-bit ADC)
3	INA219 voltage sensor
4	KY-003 Hall-effect sensor
5	ACS712 30A current sensor
6	Arduino Uno
7	Radiolink Pixroc 2.4G
8	3D-printed test bench

For the design of the electrical circuit, the following instrumentation was adopted:

- **Power system:** A 4-cell LiPo battery (14.8V, 5200mAh) connected via an XT60 plug to a power distribution board; A laptop that powered the Arduino Uno via a USB cable;

- **Autopilot:** Pixhawk 2.4.8 flight controller used for arming and signal routing;
- **Data acquisition:** Arduino Uno used for sensor reading and data logging;
- **Propeller-Motor-ESC:** One Holybro 920KV brushless motor connected through a 20A ESC and a Holybro 260 x 30 mm propeller;
- **Thrust measurement:** YZC-133 load cell with a 10 kg capacity, connected to an HX711 signal amplifier;
- **Rotational speed measurement:** Hall effect sensor KY-003 and a small magnet;
- **Electrical measurements:** INA219 module for voltage measurement and a ACS712 module for current measurement;
- **Control and telemetry:** The RadioMaster Boxer (ELRS) transmitter and FrSky X8R receiver used for throttle control; Holybro telemetry radio link used to communicate with the Pixhawk from the laptop;
- **Safety features:** A Safety switch with a buzzer, required to arm the motor; and a GPS module connected to the Pixhawk for positioning reference.

The electrical schematic shown in Figure 4 indicates that the LiPo battery is connected to the power distribution board (PDB) via an XT60 connector, and the PDB supplies the ESC, which drives the Holybro 920 KV brushless motor mounted on the test bench. In parallel, the battery voltage is stepped down by a dedicated DC-DC regulator (BEC), and the resulting regulated 5 V output powers the Pixhawk, which interfaces with the Holybro telemetry radio, the FrSky X8R receiver, the GPS module, the safety switch, and the buzzer.

A Hall effect sensor (KY-003) [8] and a small magnet attached to the motor surface were also used to measure the rotational speed (RPM). As the motor rotates, the magnet passes in front of the sensor once per revolution, generating a digital pulse (*pulses per revolution*, PPR = 1). Therefore, the rotational speed was calculated from the time interval

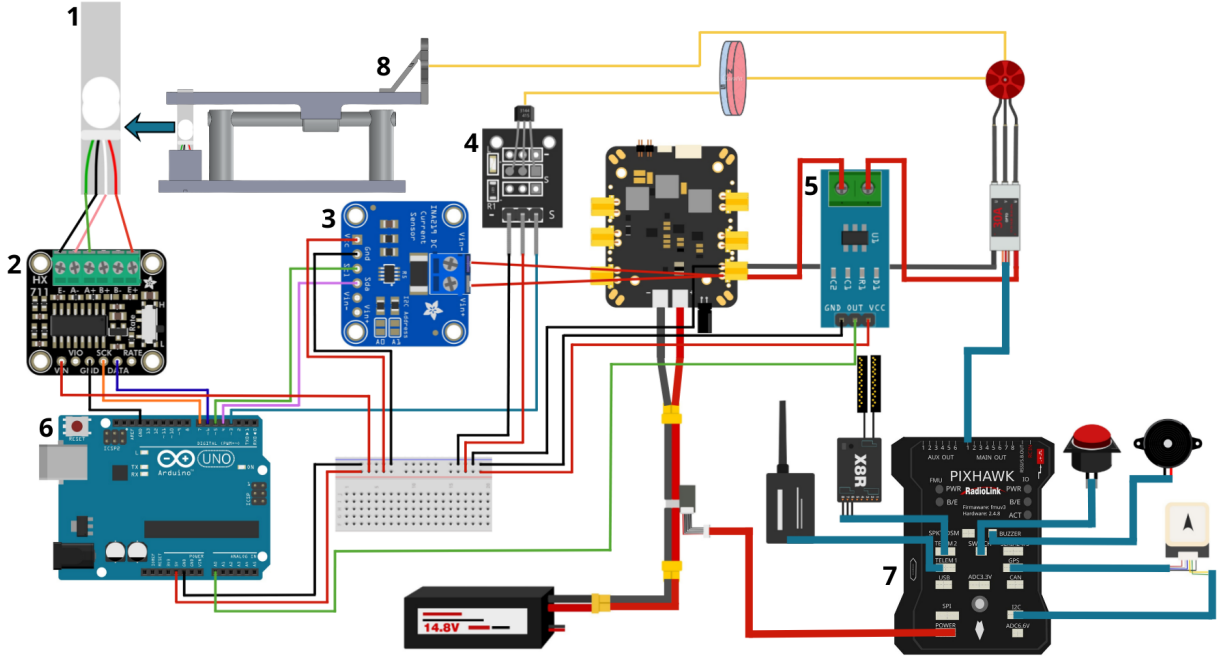


Fig. 4: Instrumentation schematic of the test bench. Thrust is acquired from a load cell via an HX711 module, RPM is measured with a KY-003 Hall-effect sensor, and electrical variables are measured with INA219 (voltage) and ACS712-30A (current), all read by an Arduino Uno. The principal components are identified from 1 to 8 at the Table I.

between two consecutive pulses, Δt , according to:

$$\text{RPM} = \frac{60}{\Delta t \cdot \text{PPR}},$$

where Δt is measured in seconds.

For thrust measurement as a function of RPM, the load cell was coupled to the bench structure, connected to the HX711 module, and the reading was taken using an Arduino UNO. However, since the HX711 reading is obtained in mass equivalent units after calibration, the thrust was converted to newtons by

$$T = m g_0,$$

where m is the estimated equivalent mass in kg and $g_0 = 9.81 \text{ m/s}^2$.

For voltage measurement as a function of RPM, the INA219 [9] module was used, powered by the Arduino UNO and connected via SDA and SCL interfaces. Although the INA219 can be installed in series on the positive path of the ESC to simultaneously measure current and voltage, it has a limited current measurement capacity of 3.2 A in the standard configuration, which is lower than the maximum expected current of the ESC motor assembly, approximately 17 A. Therefore, in this work, the INA219 was used only as a voltmeter. Thus, the VIN+ and VIN- terminals were connected to the positive power supply terminal of the ESC on the PDB, while the negative terminal of the PDB was kept in common with the GND of the Arduino because the INA219 measures the bus voltage relative to its ground (GND). This common ground reference is necessary for accurate voltage readings.

For measuring current as a function of RPM, the 30 A ACS712 sensor, which operates using the Hall effect, was used. The module was installed in series on the positive conductor between the PDB and the ESC so that all the current from the motor-ESC assembly passed through the sensor's internal conductor. When the current passes through this conductor, it generates a magnetic field around it, and the Hall effect element in the ACS712 detects this field. The sensor's internal circuit converts it into an analog output voltage, which is approximately linear with the current. This approach is useful because it allows for measuring high currents while maintaining a low-resistance power path and providing a simple signal that is easy for the microcontroller to read.

According to the datasheet, for $I = 0 \text{ A}$, the output has an offset close to $V_0 \approx V_{CC}/2$, and for the 30 A version, the typical sensitivity is $S \approx 66 \text{ mV/A}$ [10]. Thus, the current was calculated by $I = \frac{V_{\text{out}} - V_0}{S}$.

In practice, V_0 was obtained experimentally with the motor off ($I \approx 0 \text{ A}$), since this offset can vary between modules. The voltage V_{out} was read by an analog input of the Arduino UNO. Finally, the GND of the ACS712 was kept common with the GND of the Arduino and the negative terminal of the power system, ensuring a single reference for reading V_{out} .

During testing, the Arduino transmitted the measurements as simple numerical lines separated by commas. Figure 5 shows a small example of this output. Each line contains four values, in the following order: RPM, voltage, current, and thrust. The lines do not include text markers because MATLAB reads the serial stream by separating the commas.

The sensor data acquisition codes were developed in the Arduino IDE software, which is compatible with the Arduino UNO microcontroller. During testing, the microcontroller read the sensors and transmitted the data via the serial interface using USB. To monitor the results during the tests, MATLAB software was used to receive the readings and update the curves in real time. This allowed for immediate verification of the consistency of the obtained values and the identification of possible assembly errors in the electrical circuit. After the collection of the data, it was processed in MATLAB with a low-pass filter to reduce measurement noise and obtain more stable curves.

```

1500,16.26,1.4,0.38
1759,16.20,3.1,0.34
1937,16.06,3.2,0.24
2243,16.03,4.4,0.40
2401,16.11,4.5,0.46
2690,16.06,5.9,0.98
2969,15.90,6.9,1.16
3213,15.90,8.3,1.57
3395,15.82,8.7,1.83
3569,15.87,8.8,1.80
3860,15.87,10.3,2.60
4156,15.71,11.4,3.26
4311,15.72,12.7,3.54
4652,15.67,13.1,4.32
4877,15.77,13.8,4.76

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Fig. 5: Hypothetical example of the output stream used as input to MATLAB. Each row is a comma-separated numeric frame in the following order: RPM, voltage (V), current (A), and thrust (N).

III. RESULTS AND DISCUSSION

A. Structure Simulation Analysis and Total Cost Summary

In the static finite element analysis of the motor mount, the highest von Mises equivalent stress was observed in the region of the mounting hole, as indicated by the ‘max’ in Figure 3, reaching 8.2 MPa. This value remains significantly below the typical mechanical strength of PLA, which is approximately 60 MPa [11], indicating an adequate structural margin for the force considered in the simulation. Figure 3 also shows the simulation result, in which the strain field was amplified approximately 62 times solely to facilitate visualization of the deformation.

Considering the results of the structural simulation of the motor support, the test bench was then manufactured and assembled for experimental testing. The structure was produced using 3D printing, and the main components were installed, such as the linear bearings, the load cell, and the motor-ESC assembly. Figure 2 shows the assembled test bench, ready for testing.

Table II presents the final cost of the developed test bench, considering the values effectively paid for the components and manufacturing. All costs are expressed in Brazilian reais (R\$), since all components were purchased in Brazil. The total cost can be considered low for an experimental bench of this type, especially when compared with higher-precision commercial sensors, whose individual cost may exceed the total cost of the entire developed bench.

B. Performance Analysis

Among the experimental results obtained, the relationship between thrust and RPM stands out. Figure 6 shows the experimental results collected during the bench tests in blue and the reference points extracted from the manufacturer’s report in orange.

For a comparison between the two sets of data, a quadratic model based on the theory of propellers in a static regime was used, according to Leishman [12]. Starting from the expression

$$T \propto n^2$$

where n is the rotation in revolutions per second, and since the same propeller was used under constant conditions, it can be concluded that the thrust is proportional to the square of the rotation, represented by

$$T = k \cdot RPM^2$$

where T is the thrust in grams and k is a constant resulting from the experimental data. Figure 7 shows the resulting model curves for the collected data and for the manufacturer’s data. The coefficients obtained were $k_{bench} = 1.3866 \cdot 10^{-5} g/RPM^2$ and $k_{fab} = 1.3749 \cdot 10^{-5} g/RPM^2$, corresponding to a relative difference of approximately 0.85%. It is important to emphasize that the value of k_{bench} was obtained by least squares, using all 954 pairs of measured points, seeking an estimate as close to the real value as possible. Furthermore, when comparing the collected data with the manufacturer’s curve, the coefficient of determination of $r^2 = 0.9916$ was obtained, indicating good agreement between the test measurements and the manufacturers’ reference values for the Holybro motor-propeller assembly analyzed.

TABLE II: Acquisition and manufacturing costs of the developed test bench

Item	Cost (R\$)
ACS712 current sensor	14.90
INA219 voltage and current sensor	14.56
KY-003 Hall effect sensor	13.90
10 kg load cell kit with HX711 amplifier	33.90
Arduino Uno R3	49.00
Breadboard with jumper wires	22.90
Set of 6 alligator clips with thick wires	20.00
Linear bearings with linear shafts	49.90
Approximate 3D printing and energy cost	20.00
Screws, nuts, and washers	10.00
Total	249.06

For the Voltage vs RPM curve, shown in Figure 8, the experimental data, represented in blue, indicate an almost constant voltage throughout the test, with a slight drop as RPM increases. This trend is expected: as the system current demand rises and the required power increases, the battery voltage decreases under load, mainly due to the behavior of internal resistance and circuit losses that are characteristic of Li-Po batteries.

In contrast, the manufacturer's reference data, represented by orange markers, remain constant at 16 V over the entire available RPM range. A plausible explanation is that the manufacturer performed the tests using a regulated power supply in order to keep the input voltage fixed. Therefore, the comparison between the measured data and the manufacturer's data indicates that the observed discrepancy is not due to measurement error but rather to different power-supply

conditions during testing. In addition, the difference between the measured voltage and the 16 V reference remains small, with a maximum deviation on the order of 0.5 V, which suggests good consistency in the measured results.

For the current measurement test as a function of RPM, Figure 9 shows the data obtained on the test bench in blue and the reference points provided by the manufacturer in orange. A good agreement is observed in the data obtained from the tests, indicating that the experimental results follow the expected trend.

The relationship between current and rotation is expected to exhibit cubic behavior since, in Leishman propeller theory [12], the absorbed power can be expressed by the dimensionless power coefficient $C_p = P/(\rho n^2 D^5)$, and, for the same propeller operating under constant ρ conditions, it is obtained that $P \propto n^3$ with n in revolutions per second. Therefore,

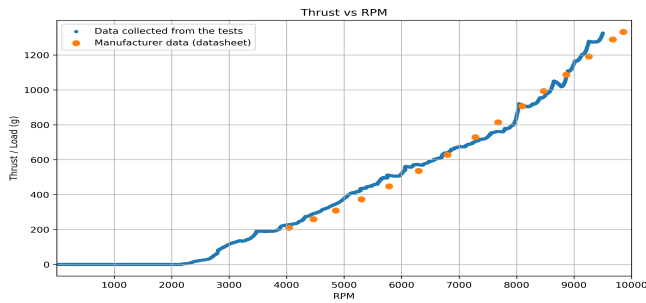


Fig. 6: Thrust (g) vs RPM for experimental measurements from the test bench compared with the manufacturer's reference data

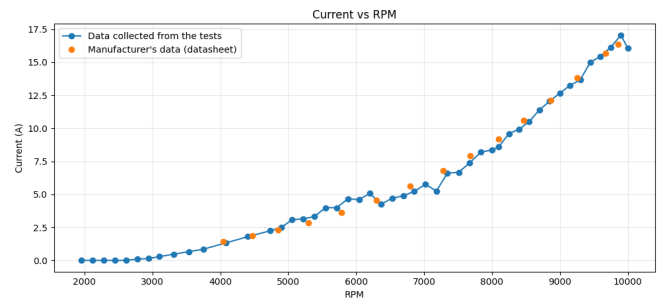


Fig. 9: Current (A) vs RPM for the experimental measurements compared with the manufacturer's reference data

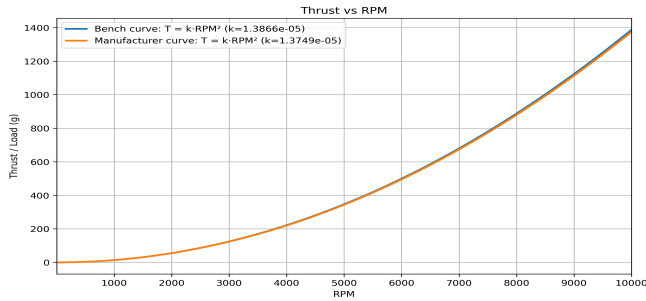


Fig. 7: Thrust (g) vs RPM: Curve fitting for experimental data and manufacturer's data

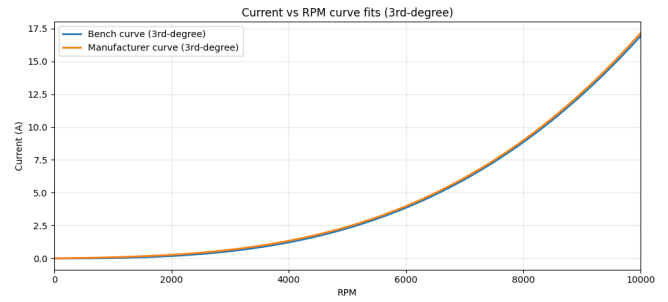


Fig. 10: Current (A) vs RPM: Curve fitting for experimental data and manufacturer's data

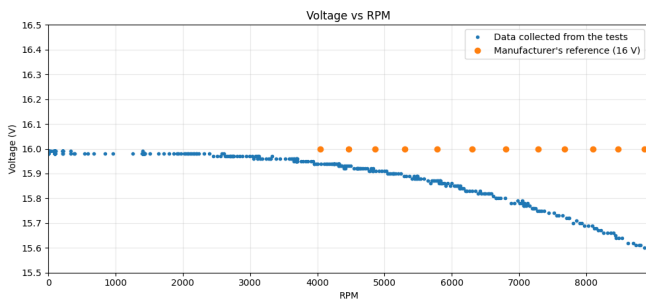


Fig. 8: Voltage (V) vs RPM for experimental measurements compared with the manufacturer's reference data

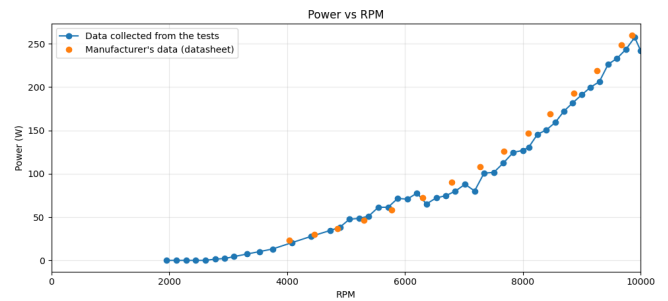


Fig. 11: Power (W) vs RPM for experimental measurements from the test bench compared with the manufacturer's reference data

since $P_{el} = V \cdot I$ is under approximately constant supply voltage, it follows that $I \propto n^3$. Consequently, $I \propto RPM^3$, justifying the use of a third-degree curve fit to represent the trend of the data, as shown in Figure 10. The coefficient of determination of $r^2 = 0.9988$ was obtained, highlighting the close proximity between the curves.

Based on the measured voltage and current data, it was also possible to estimate the electrical power required by the Holybro motor-ESC-propeller assembly tested under each rotational condition. Considering electrical power as $P = V \cdot I$, where P was calculated by taking the voltage and current values, both at the same RPM. Figure 11 shows the Power vs RPM curve obtained during the tests, along with the manufacturer's reference, thus allowing the consistency of the system's energy behavior to be evaluated.

Since the voltage and current measurements showed good agreement with the manufacturer's data, the estimated power also proved to be compatible. Furthermore, it is noticeable that since the voltage remains practically constant, undergoing only small variations, the graph presents a curve similar to that in Figure 9. Thus, the results of the graph indicate that the experimental procedure was able to consistently capture the power required throughout the operating range.

Experimental validation demonstrated that the proposed test bench is capable of providing reliable propulsion data. The relationships between thrust, current, voltage, and rotational speed obtained during the experiments showed consistent behavior when compared with the manufacturer datasheet of the Holybro motor-propeller assembly, confirming the accuracy of the instrumentation and the adequacy of the measurement approach. These results indicate that the developed platform can effectively reproduce the expected performance characteristics of UAV propulsion systems under static testing conditions.

IV. CONCLUSION AND FUTURE WORK

This work presented the design, implementation, and experimental validation of a low-cost static test bench for brushless motors used in UAV propulsion systems. Experimental validation confirmed that the developed test bench is capable of providing reliable data. The thrust, current, and voltage curves relative to RPM obtained in the tests showed consistent behavior with the data from the manufacturer's datasheet, Holybro, reinforcing the functionality of the instrumentation used. In addition to adequate performance, another purpose of the test bench was accessibility, such as the use of low-cost components and simple assembly, making the bench a viable alternative not only for the UFVision extension project team but also for other university teams and small businesses.

Additionally, the validated system of the test bench has the capacity to compare different propulsion configurations, allowing for the evaluation of variations in propellers, motors, ESCs, and power supply conditions, such as batteries and power sources.

For future work, a simpler assembly system is planned to streamline the alignment of all components and sensors, as this step proved to be one of the most time-consuming during development. In addition, a method will be implemented to measure and evaluate the generated torque, allowing for a more complete characterization of the propulsion system.

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HARDWARE AVAILABILITY

The hardware files related to the developed test bench are publicly available in a GitHub repository (link to hardware files). The repository currently includes only the mechanical design files, such as SolidWorks and STEP files.

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